

2024 AMC 10B Problem 9 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10B Problem 9. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10B Problem 9

**Problem:**

Real numbers a , b , and c have arithmetic mean 0. The arithmetic mean of a^2 , b^2 , and c^2 is 10. What is the arithmetic mean of ab , ac , and bc ?

Answer Choices:

A. -5

B. $-\frac{10}{3}$

C. $-\frac{10}{9}$

D. 0

E. $\frac{10}{9}$

Solution:

The given information implies that $a + b + c = 0$ and $a^2 + b^2 + c^2 = 30$. Then

$$0 = (a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2ac + 2bc = 30 + 2(ab + ac + bc)$$

Therefore $2(ab + ac + bc) = -30$ and the requested arithmetic mean is $\frac{ab+ac+bc}{3} = \frac{-15}{3} = (A) \boxed{-5}$.

OR

Consider the system of equations implied by the conditions of the problem,

$$\begin{aligned}a + b + c &= 0 \\ a^2 + b^2 + c^2 &= 30\end{aligned}$$

and suppose that $a = 0$. Then $b + c = 0$, so $b = -c$, and substituting into the second equation gives $2b^2 = 30$, from which $b = \pm\sqrt{15}$ and $c = \mp\sqrt{15}$. If one assumes that the requested arithmetic mean is determined by the given information, independent of the value of a , then

$$\frac{ab + ac + bc}{3} = \frac{0 + 0 - \sqrt{15} \cdot \sqrt{15}}{3} = (\text{A}) \boxed{-5}$$

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